

# Graphs of parabolas

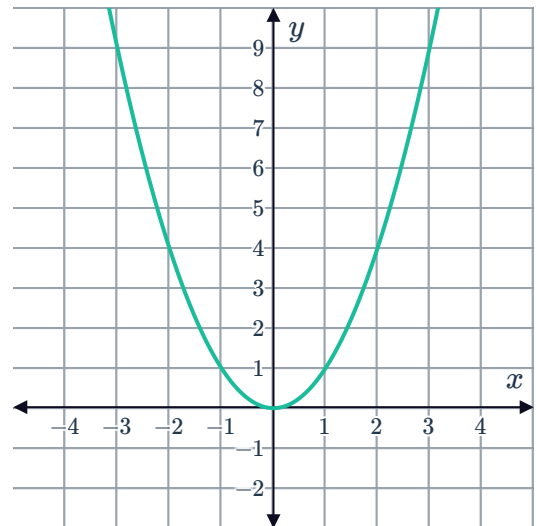


1

a

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2 | 3 |
| $y$ | 9  | 4  | 1  | 0 | 1 | 4 | 9 |

b



c No

d  $x = 0$

e 0

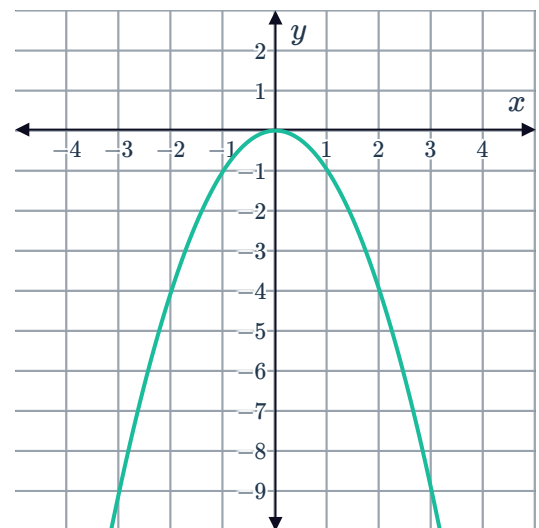
f 2

2

a

|     |    |    |    |   |    |    |    |
|-----|----|----|----|---|----|----|----|
| $x$ | -3 | -2 | -1 | 0 | 1  | 2  | 3  |
| $y$ | -9 | -4 | -1 | 0 | -1 | -4 | -9 |

b



c No

d  $x = 0$

e 0

3

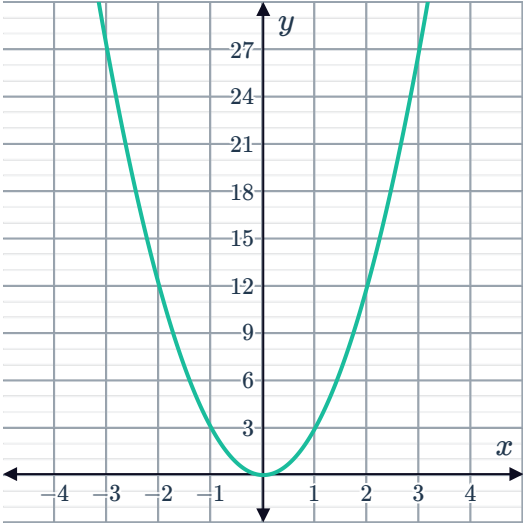
a

i



|     |    |    |    |   |   |    |    |
|-----|----|----|----|---|---|----|----|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2  | 3  |
| $y$ | 27 | 12 | 3  | 0 | 3 | 12 | 27 |

ii

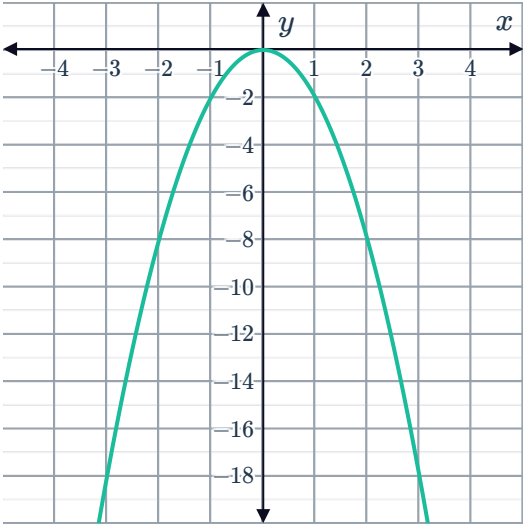


b

i

|     |     |    |    |   |    |    |     |
|-----|-----|----|----|---|----|----|-----|
| $x$ | -3  | -2 | -1 | 0 | 1  | 2  | 3   |
| $y$ | -18 | -8 | -2 | 0 | -2 | -8 | -18 |

ii

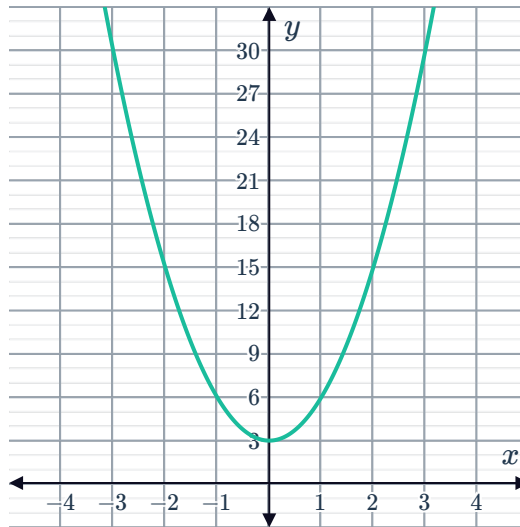


c  
i



|     |    |    |    |   |   |    |    |
|-----|----|----|----|---|---|----|----|
| $x$ | -3 | -2 | -1 | 0 | 1 | 2  | 3  |
| $y$ | 30 | 15 | 6  | 3 | 6 | 15 | 30 |

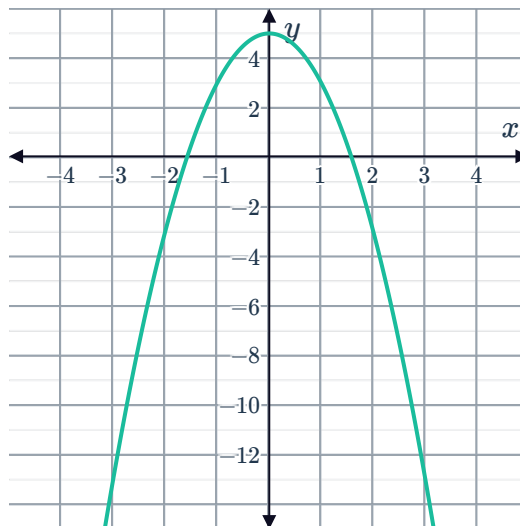
ii



d  
i

|     |     |    |    |   |   |    |     |
|-----|-----|----|----|---|---|----|-----|
| $x$ | -3  | -2 | -1 | 0 | 1 | 2  | 3   |
| $y$ | -13 | -3 | 3  | 5 | 3 | -3 | -13 |

ii

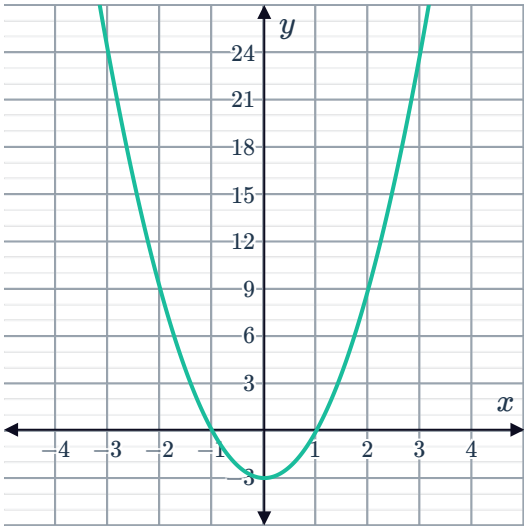


e  
i



|     |    |    |    |    |   |   |    |
|-----|----|----|----|----|---|---|----|
| $x$ | -3 | -2 | -1 | 0  | 1 | 2 | 3  |
| $y$ | 24 | 9  | 0  | -3 | 0 | 9 | 24 |

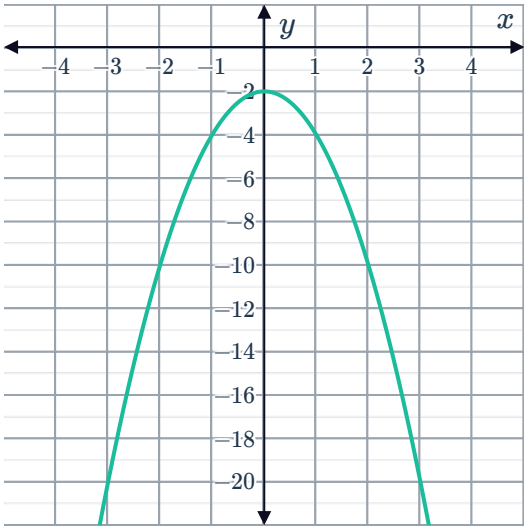
ii



f  
i

|     |     |     |    |    |    |     |     |
|-----|-----|-----|----|----|----|-----|-----|
| $x$ | -3  | -2  | -1 | 0  | 1  | 2   | 3   |
| $y$ | -20 | -10 | -4 | -2 | -4 | -10 | -20 |

ii



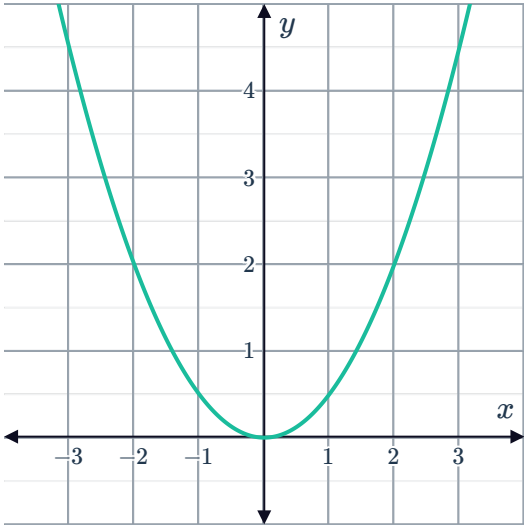
g

i



|     |               |    |               |   |               |   |               |
|-----|---------------|----|---------------|---|---------------|---|---------------|
| $x$ | -3            | -2 | -1            | 0 | 1             | 2 | 3             |
| $y$ | $\frac{9}{2}$ | 2  | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | 2 | $\frac{9}{2}$ |

ii

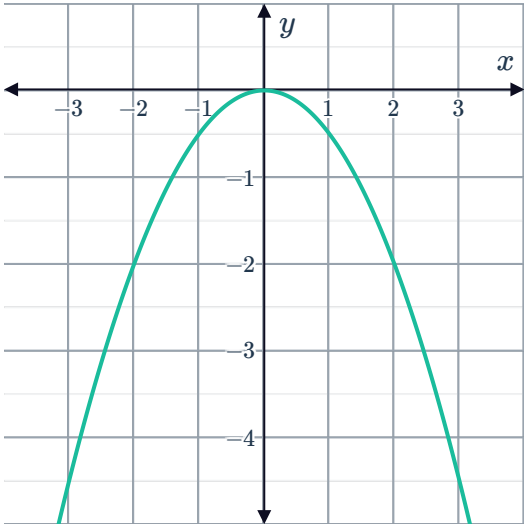


h

i

|     |                |    |                |   |                |    |                |
|-----|----------------|----|----------------|---|----------------|----|----------------|
| $x$ | -3             | -2 | -1             | 0 | 1              | 2  | 3              |
| $y$ | $-\frac{9}{2}$ | -2 | $-\frac{1}{2}$ | 0 | $-\frac{1}{2}$ | -2 | $-\frac{9}{2}$ |

ii





## Transformations of parabolas

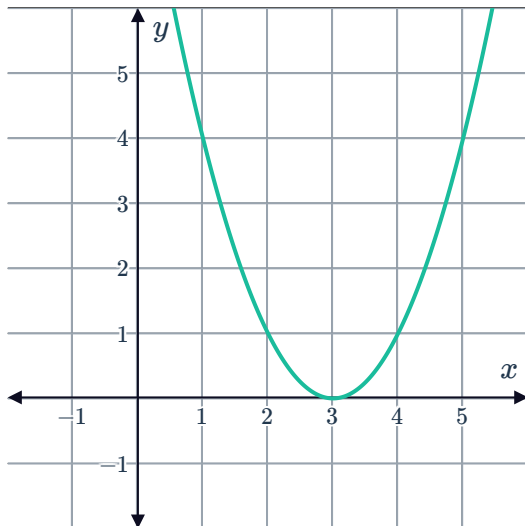
4

a

i

|     |   |   |   |   |   |
|-----|---|---|---|---|---|
| $x$ | 1 | 2 | 3 | 4 | 5 |
| $y$ | 4 | 1 | 0 | 1 | 4 |

ii



iii 0

iv  $x = 3$

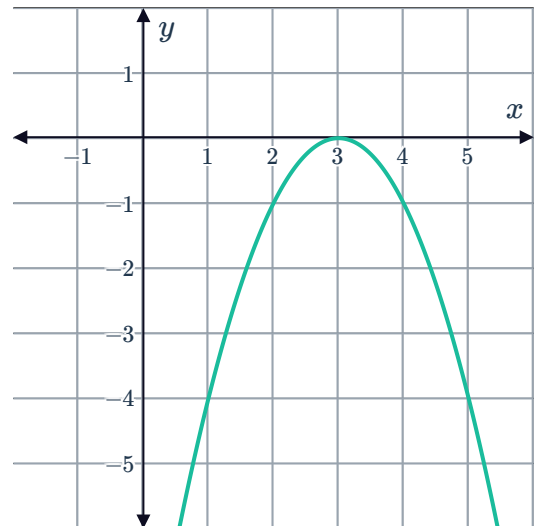
v (3, 0)

b

i

|     |    |    |   |    |    |
|-----|----|----|---|----|----|
| $x$ | 1  | 2  | 3 | 4  | 5  |
| $y$ | -4 | -1 | 0 | -1 | -4 |

ii



iii 0

iv 3

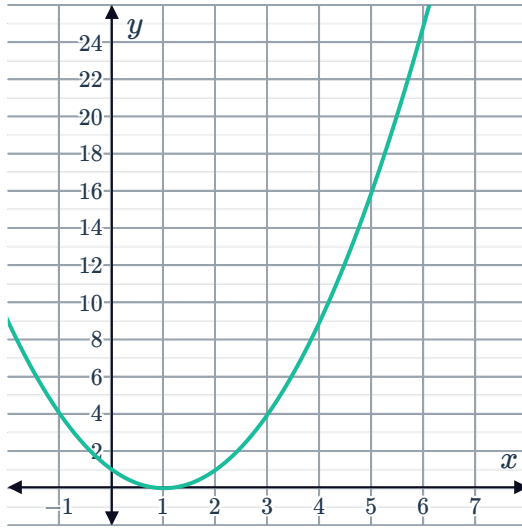
v (3, 0)

c

i

|     |   |   |   |   |    |
|-----|---|---|---|---|----|
| $x$ | 1 | 2 | 3 | 4 | 5  |
| $y$ | 0 | 1 | 4 | 9 | 16 |

ii



iii 0

iv 1

v (1, 0)

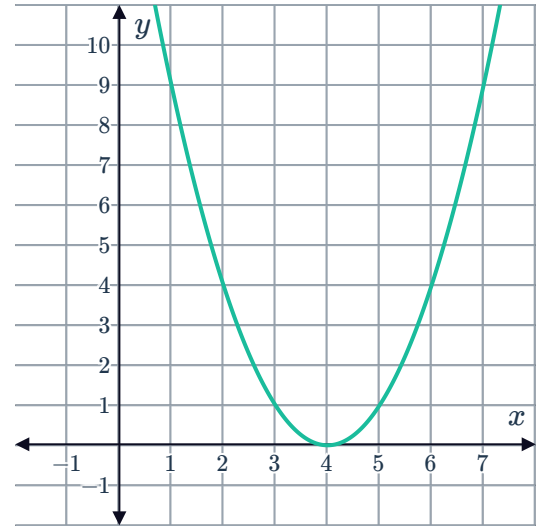


d

i

|     |   |   |   |   |   |
|-----|---|---|---|---|---|
| $x$ | 1 | 2 | 3 | 4 | 5 |
| $y$ | 9 | 4 | 1 | 0 | 1 |

ii



iii 0

iv 4

v (4, 0)

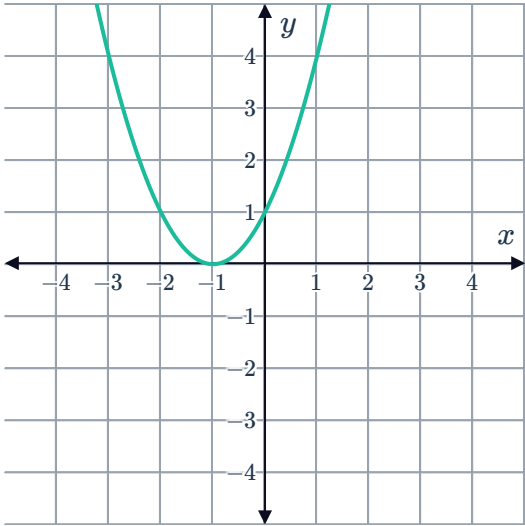
5



a  
i

|     |    |    |    |   |   |
|-----|----|----|----|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 |
| $y$ | 4  | 1  | 0  | 1 | 4 |

ii

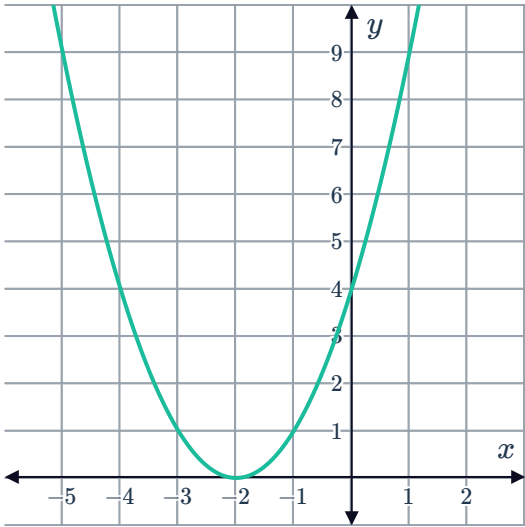


- iii 0
- iv  $x = -1$
- v  $(-1, 0)$

c  
i

|     |    |    |    |   |   |
|-----|----|----|----|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1 |
| $y$ | 1  | 0  | 1  | 4 | 9 |

ii

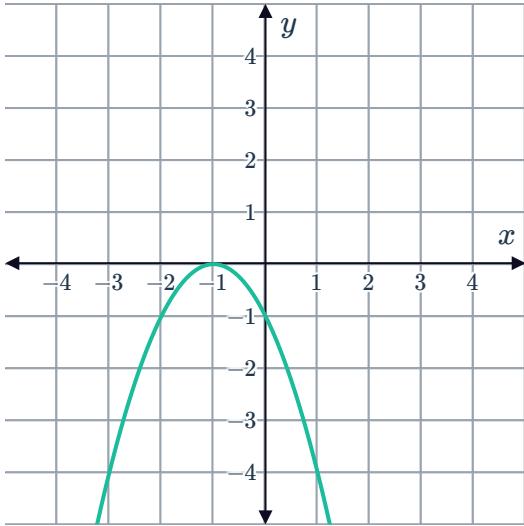


- iii 0
- iv -2
- v  $(-2, 0)$

b  
i

|     |    |    |    |    |    |
|-----|----|----|----|----|----|
| $x$ | -3 | -2 | -1 | 0  | 1  |
| $y$ | -4 | -1 | 0  | -1 | -4 |

ii

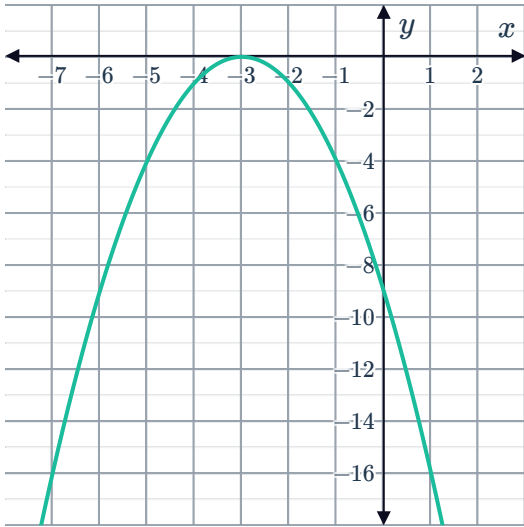


- iii 0
- iv -1
- v  $(-1, 0)$

d  
i

|     |    |    |    |    |     |
|-----|----|----|----|----|-----|
| $x$ | -3 | -2 | -1 | 0  | 1   |
| $y$ | 0  | -1 | -4 | -9 | -16 |

ii



- iii 0
- iv -3
- v  $(-3, 0)$

6

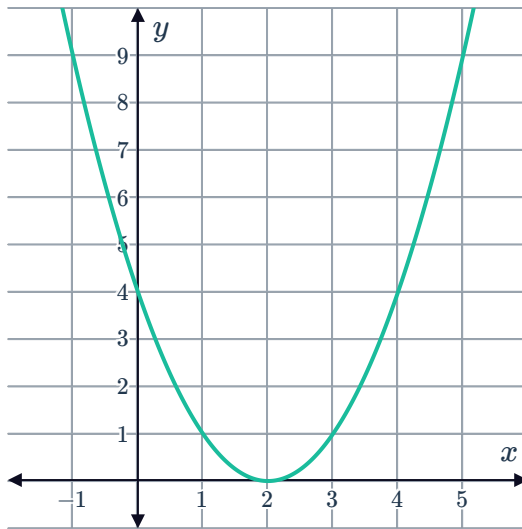


a

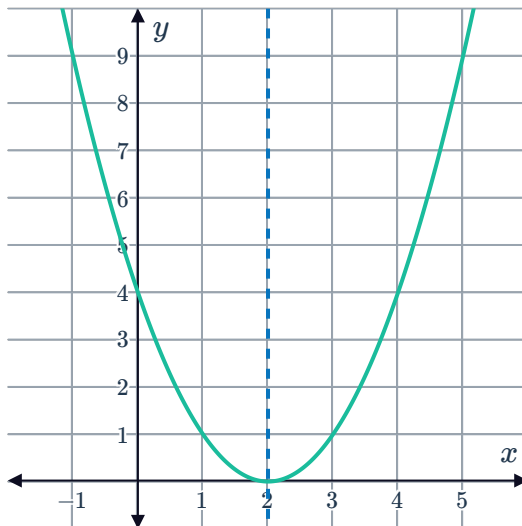
i  $(2, 0)$

ii  $x = 2$

iii



iv

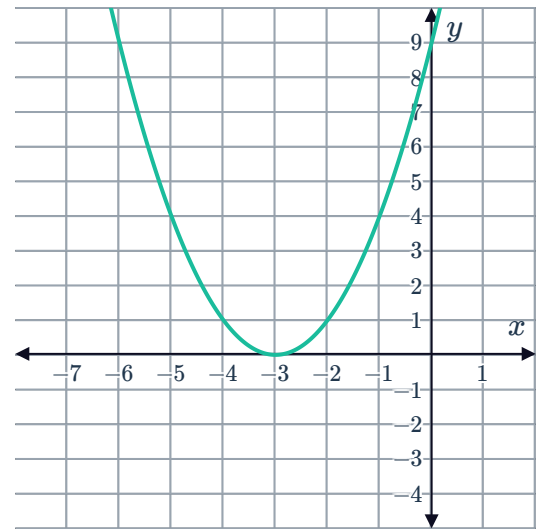


b

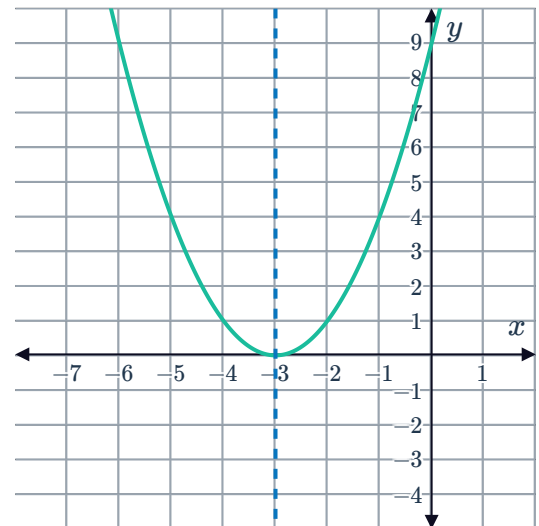
i  $(-3, 0)$

ii  $x = -3$

iii



iv

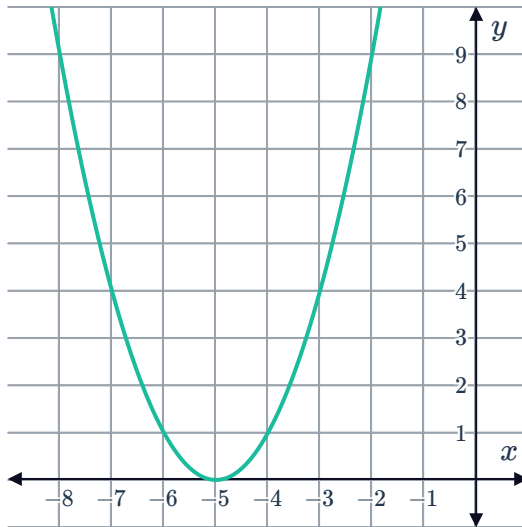


c

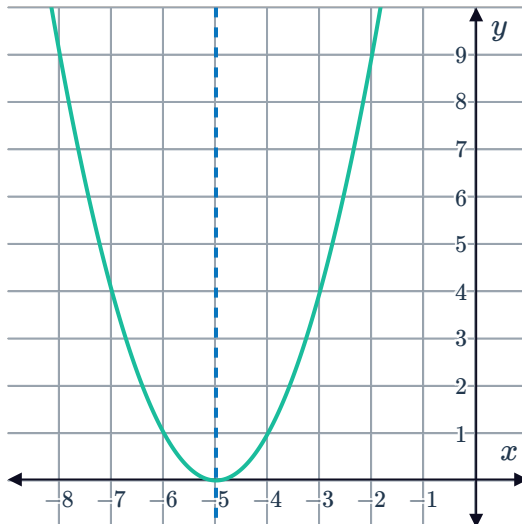
i  $(-5, 0)$

ii  $x = -5$

iii



iv

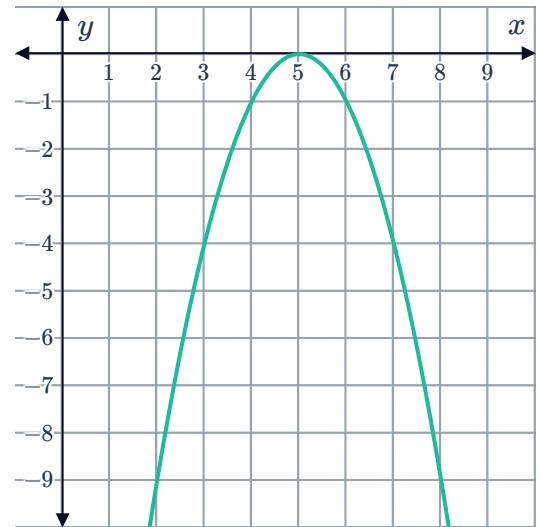


d

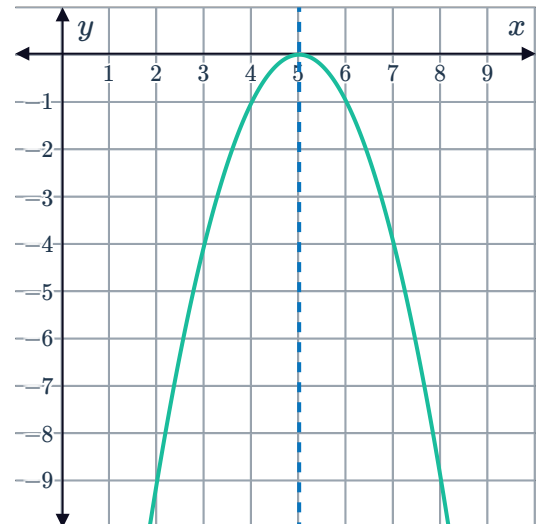
i  $(5, 0)$

ii  $x = 5$

iii



iv



7 Horizontally translated by  $-4$  units.

8

a Concave up

b Concave down

c Concave down

d Concave up

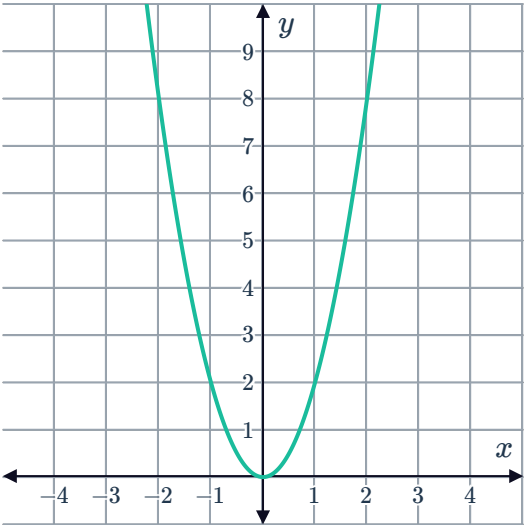
9

a

i

|     |    |    |   |   |   |
|-----|----|----|---|---|---|
| $x$ | -2 | -1 | 0 | 1 | 2 |
| $y$ | 8  | 2  | 0 | 2 | 8 |

ii



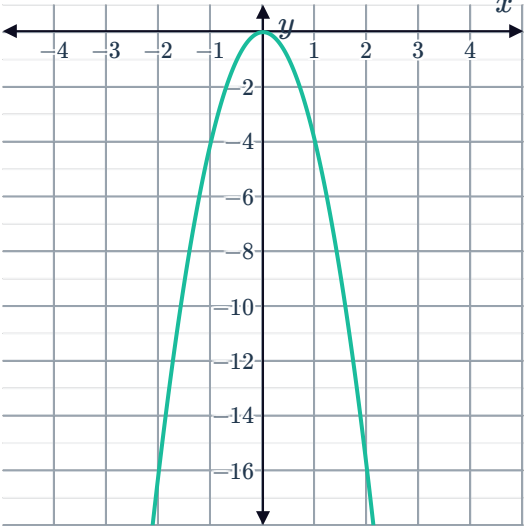
iii (0,0)

c

i

|     |     |    |   |    |     |
|-----|-----|----|---|----|-----|
| $x$ | -2  | -1 | 0 | 1  | 2   |
| $y$ | -16 | -4 | 0 | -4 | -16 |

ii



iii (0,0)

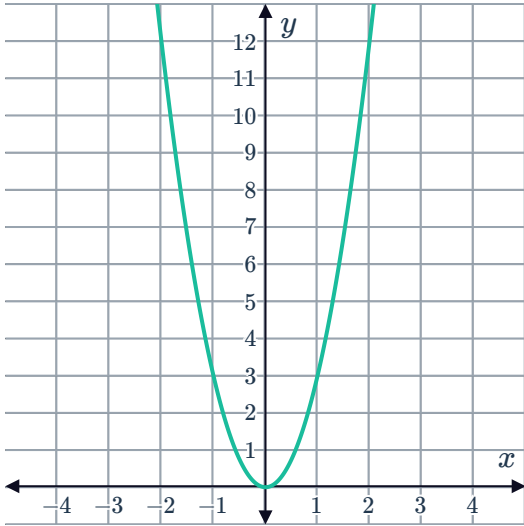


b

i

|     |    |    |   |   |    |
|-----|----|----|---|---|----|
| $x$ | -2 | -1 | 0 | 1 | 2  |
| $y$ | 12 | 3  | 0 | 3 | 12 |

ii



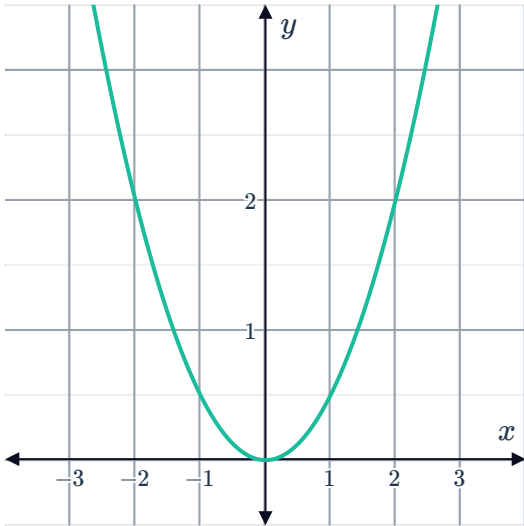
iii (0,0)

d

i

|     |    |               |   |               |   |
|-----|----|---------------|---|---------------|---|
| $x$ | -2 | -1            | 0 | 1             | 2 |
| $y$ | 2  | $\frac{1}{2}$ | 0 | $\frac{1}{2}$ | 2 |

ii



iii (0,0)

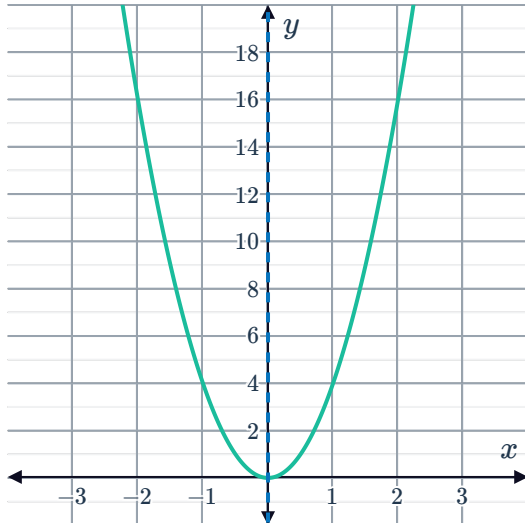
10



a

i  $x = 0$

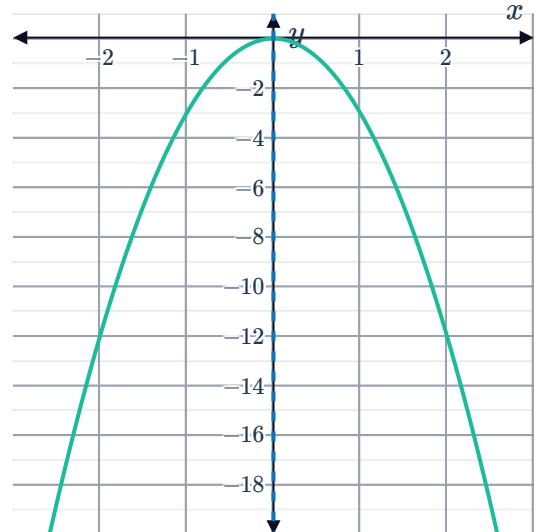
ii



b

i  $x = 0$

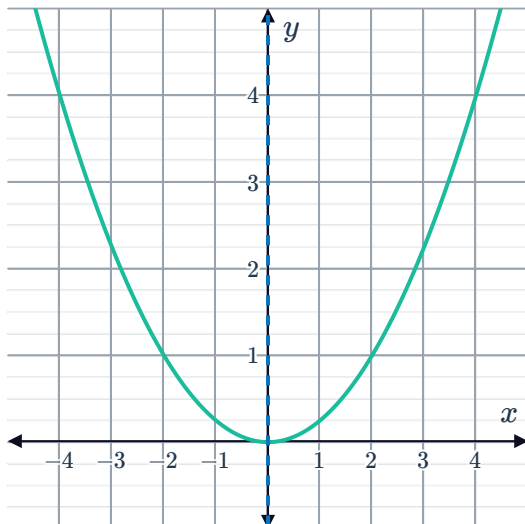
ii



c

i  $x = 0$

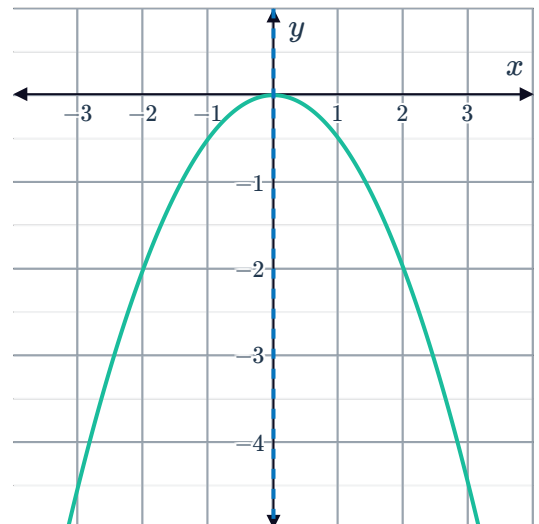
ii



d

i  $x = 0$

ii



11 Vertically scaled by 5 units.

12

a  $y = x^2 + 3$

b  $y = x^2 - 2$

c  $y = (x - 5)^2$

d  $y = (x + 4)^2$

e  $y = 2x^2$

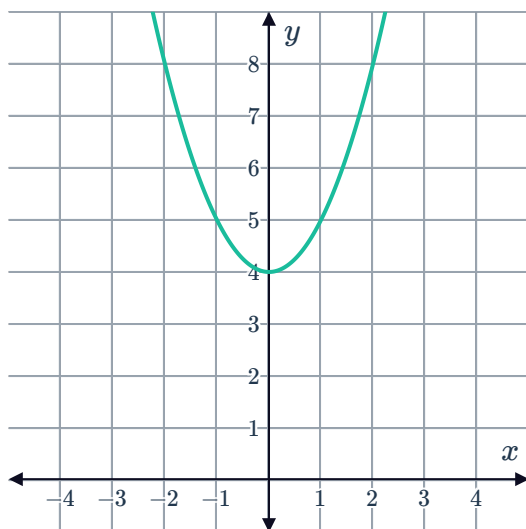
f  $y = -3x^2$

g  $y = \frac{1}{2}x^2$

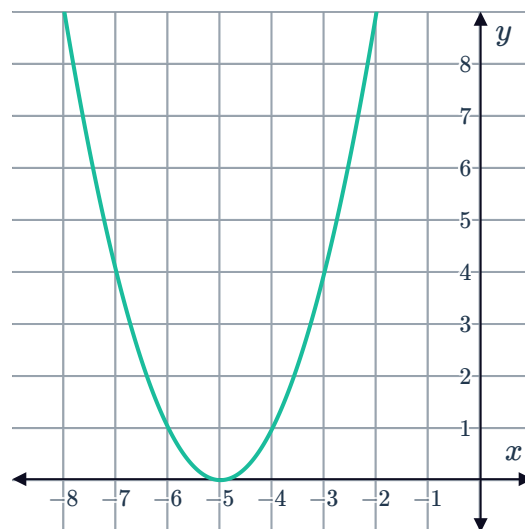
h  $y = -x^2$

13

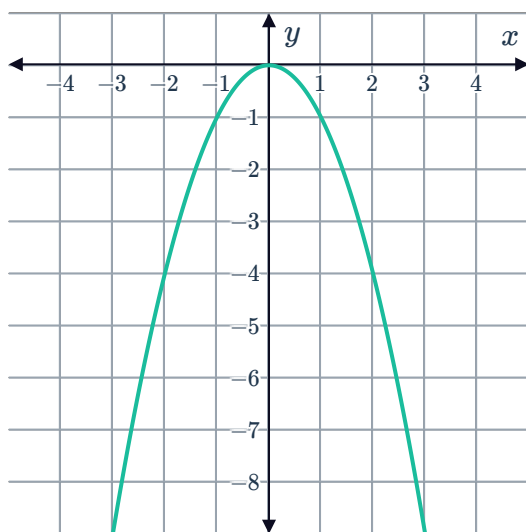
a



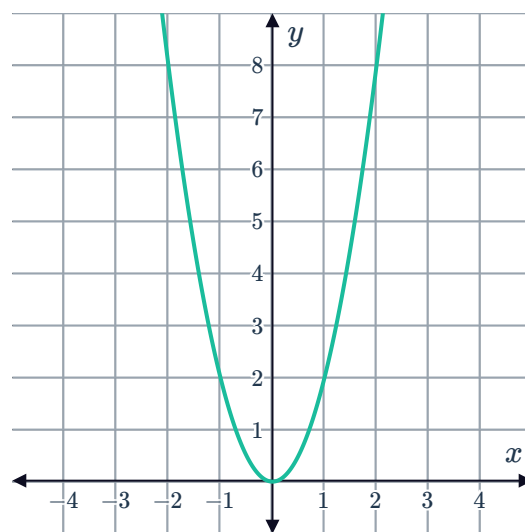
b



c



d



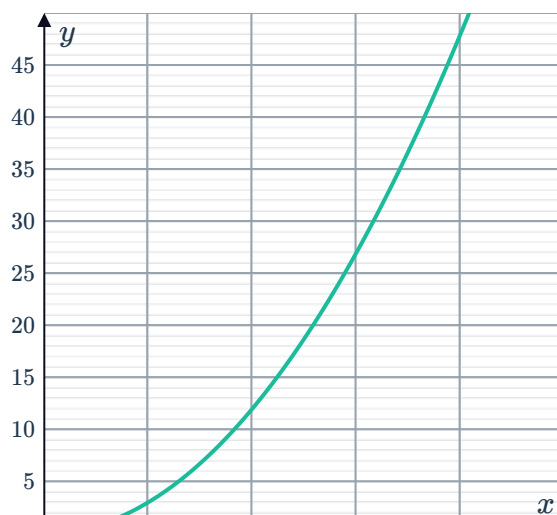
## Applications

14

a

|         |   |   |    |    |    |
|---------|---|---|----|----|----|
| $x$ (m) | 0 | 1 | 2  | 3  | 4  |
| $y$ (m) | 0 | 3 | 12 | 27 | 48 |

b



c 0 m

e (0,0)



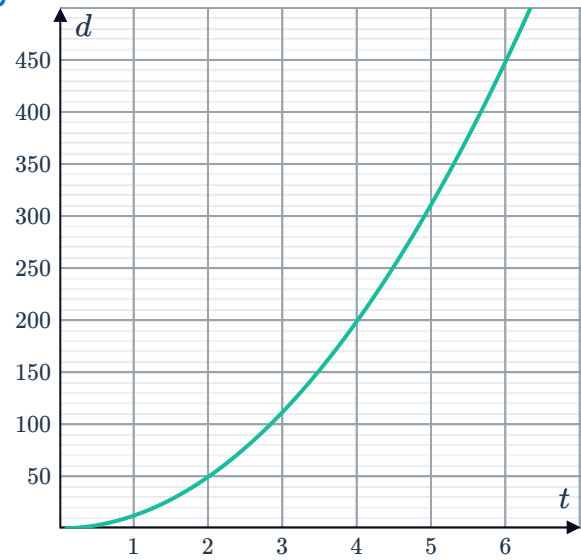
d  $x = 0$

15

a

|                  |   |    |     |     |
|------------------|---|----|-----|-----|
| time ( $t$ )     | 0 | 2  | 4   | 6   |
| distance ( $d$ ) | 0 | 50 | 200 | 450 |

b



c  $t = 3$  s